Beyond Self-Attention: Deformable Large Kernel Attention for Medical Image Segmentation

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论文地址：https://arxiv.org/pdf/2309.00121

Github 地址：https://github.com/xmindflow/deformableLKA

这是 WACV2024 的一篇文章。

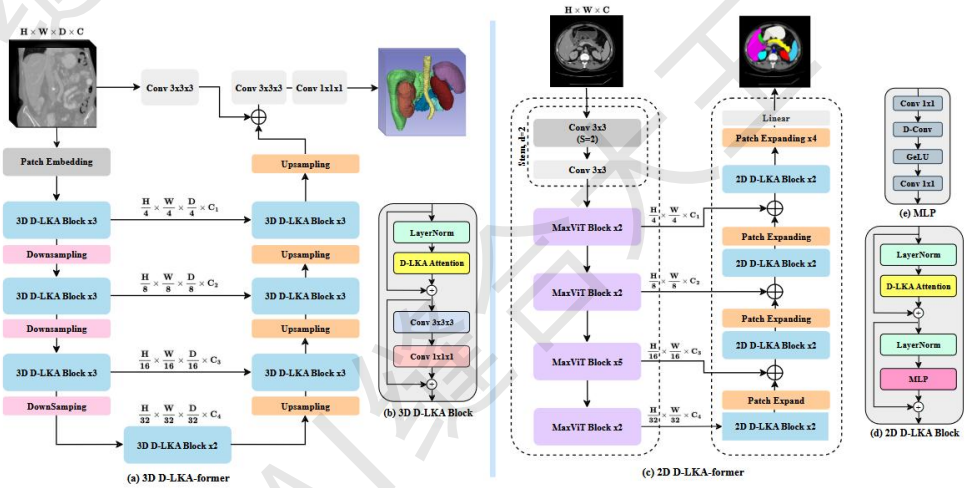
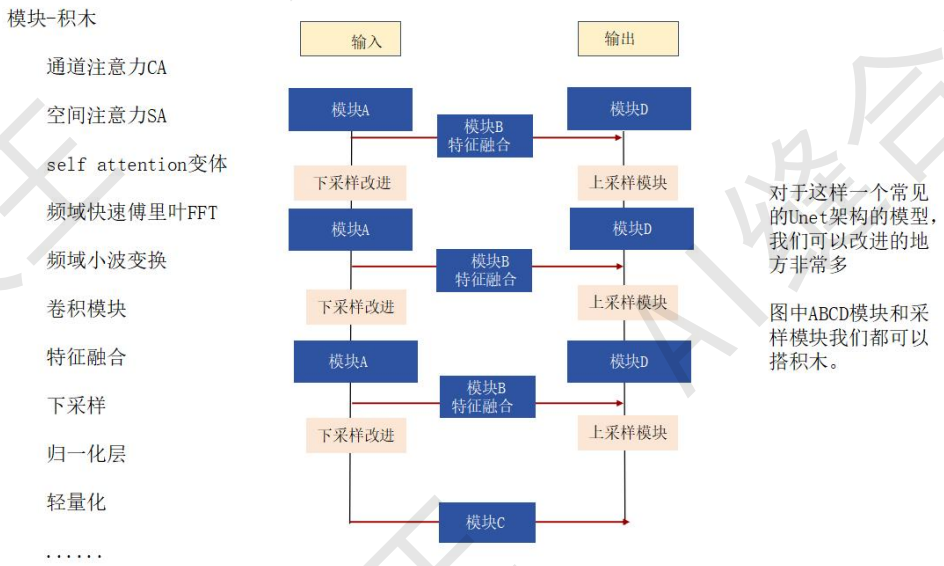


Figure 1. Proposed network architecture of the 3D D-LKA model on the left and the 2D D-LKA model on the right.



根据之前发的这个图可以看到，这篇论文提出的模型是在模块 A 部分使用了别人的模

块(MaxVIT)，在模块 D 部分使用了自己提出的模块(D-LKA)，可以看到 D-LKA 也是 xx-former 结构，修改 token mixer 和 ffn.

(LKA 是 22 年的一篇文章(https://arxiv.org/pdf/2202.09741))

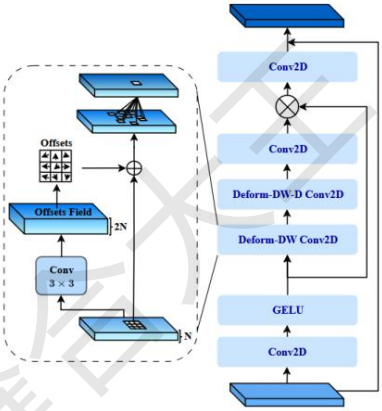
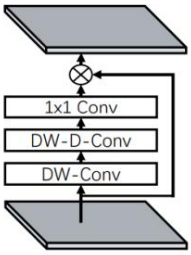


Figure 2. Architecture of the deformable LKA module.



(a) LKA

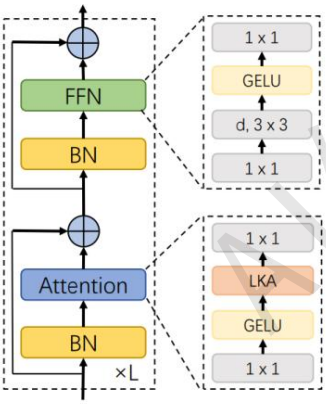


Fig. 4. A stage of VAN. d means depth wise convolution. k × k denotes k × k convolution.

可以看到论文提出的模块 D-LKA 跟 LKA 结构是一样的，只不过把 DW-Conv 深度卷积跟 DW-D-Conv 膨胀卷积，变成了可变性卷积。

所以这篇文章，其实就在 LKA 模块上应用了一下可变性卷积，实验结果跟 LKA 相比，效果有所提升。

Table 1. Comparison results of the proposed method evaluated on the Synapse dataset. Blue indicates the best result, and red displays the second-best. Parameters are reported in millions (M) and FLOPS in billions (G). DSC is presented for abdominal organs spleen (Spl), right kidney (RKid), left kidney (LKid), gallbladder (Gal), liver (Liv), stomach (Sto), aorta (Aor), and pancreas (Pan).

Methods	Params (M)	FLOPs (G)	Spl	RKid	LKid	Gal	Liv	Sto	Aor	Pan	Average	
											DSC ↑	HD95 ↓
TransUNet [15]	96.07	88.91	85.08	77.02	81.87	63.16	94.08	75.62	87.23	55.86	77.49	31.69
Swin-UNet [11]	27.17	6.16	90.66	79.61	83.28	66.53	94.29	76.60	85.47	56.58	79.13	21.55
LeViT-UNet-384 [62]	52.17	25.55	88.86	80.25	84.61	62.23	93.11	72.76	87.33	59.07	78.53	16.84
MISSFormer [33]	42.46	9.89	91.92	82.00	85.21	68.65	94.41	80.81	86.99	65.67	81.96	18.20
ScaleFormer [31]	111.6	48.93	89.40	83.31	86.36	74.97	95.12	80.14	88.73	64.85	82.86	16.81
HiFormer-B [30]	25.51	8.045	90.99	79.77	85.23	65.23	94.61	81.08	86.21	59.52	80.39	14.70
DAEFormer [4]	48.07	27.89	91.82	82.39	87.66	71.65	95.08	80.77	87.84	63.93	82.63	16.39
TransDeepLab [7]	21.14	16.31	89.00	79.88	84.08	69.16	93.53	78.40	86.04	61.19	80.16	21.25
PVT-CASCADE [47]	35.28	6.40	90.1	80.37	82.23	70.59	94.08	83.69	83.01	64.43	81.06	20.23
LKA Baseline	85.82	13.62	91.45	81.93	84.93	71.05	94.87	83.71	87.48	66.76	82.77	17.42
2D D-LKA Net	101.64	19.92	91.22	84.92	88.38	73.79	94.88	84.94	88.34	67.71	84.27	20.04

Table 2. Performance comparison of the proposed method against the SOTA approaches on skin lesion segmentation benchmarks.

Methods	ISIC 2017				ISIC 2018				PH^2			
	DSC	SE	SP	ACC	DSC	SE	SP	ACC	DSC	SE	SP	ACC
U-Net [48]	0.8159	0.8172	0.9680	0.9164	0.8545	0.8800	0.9697	0.9404	0.8936	0.9125	0.9588	0.9233
Att-UNet [50]	0.8082	0.7998	0.9776	0.9145	0.8566	0.8674	0.9863	0.9376	0.9003	0.9205	0.9640	0.9276
DAGAN [39]	0.8425	0.8363	0.9716	0.9304	0.8807	0.9072	0.9588	0.9324	0.9201	0.8320	0.9640	0.9425
TransUNet [15]	0.8123	0.8263	0.9577	0.9207	0.8499	0.8578	0.9653	0.9452	0.8840	0.9063	0.9427	0.9200
MCGU-Net [2]	0.8927	0.8502	0.9855	0.9570	0.8950	0.8480	0.9860	0.9550	0.9263	0.8322	0.9714	0.9537
MedT [55]	0.8037	0.8064	0.9546	0.9090	0.8389	0.8252	0.9637	0.9358	0.9122	0.8472	0.9657	0.9416
FAT-Net [58]	0.8500	0.8392	0.9725	0.9326	0.8903	0.9100	0.9699	0.9578	0.9440	0.9441	0.9741	0.9703
TMU-Net [8]	0.9164	0.9128	0.9789	0.9660	0.9059	0.9038	0.9746	0.9603	0.9414	0.9395	0.9756	0.9647
Swin-Unet [11]	0.9183	0.9142	0.9798	0.9701	0.8946	0.9056	0.9798	0.9645	0.9449	0.9410	0.9564	0.9678
DeepLabv3+ (CNN) [17]	0.9162	0.8733	0.9921	0.9691	0.8820	0.8560	0.9770	0.9510	0.9202	0.8818	0.9832	0.9503
HiFormer-B [29]	0.9253	0.9155	0.9840	0.9702	0.9102	0.9119	0.9755	0.9621	0.9460	0.9420	0.9772	0.9661
LKA (Baseline)	0.9229	0.9192	0.9824	0.9704	0.9118	0.8984	0.9802	0.9632	0.9409	0.9312	0.9759	0.9612
2D D-LKA Net	0.9254	0.9327	0.9793	0.9705	0.9177	0.9164	0.9773	0.9647	0.9490	0.9430	0.9775	0.9659

Table 3. Results on the Synapse dataset using 3D models. Blue indicates the best result, and red displays the second-best.

Methods	Params (M)	FLOPs (G)	Spl	RKid	LKid	Gal	Liv	Sto	Aor	Pan	Average	
											DSC ↑	HD95 ↓
UNETR [26]	92.49	75.76	85.00	84.52	85.60	56.30	94.57	70.46	89.80	60.47	78.35	18.59
Swin-UNETR [25]	62.83	384.2	95.37	86.26	86.99	66.54	95.72	77.01	91.12	68.80	83.48	10.55
nnFormer [64]	150.5	213.4	90.51	86.25	86.57	70.17	96.84	86.83	92.04	83.35	86.57	10.63
UNETR++ [51]	42.96	47.98	95.77	87.18	87.54	71.25	96.42	86.01	92.52	81.10	87.22	7.53
LKA Baseline	28.94	48.79	90.49	87.54	87.57	63.81	96.96	84.89	93.22	82.46	85.87	14.35
D-LKA Net	42.35	66.96	95.88	88.50	87.64	72.14	96.25	85.03	92.87	81.64	87.49	9.57